

Group 6 - Final Report

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To start the code, the following packages should be installed.

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)
```

```
library(Hmisc, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(table1, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(readr, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(mice, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(dplyr, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(tidyselect, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(DataExplorer, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(ggplot2, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(reshape2, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
library(VIM, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
```

Introduction

Data Source

The laptop dataset is from Kaggle [Laptop Sales Price Prediction](#).

```
laptop <- read.csv("C:/github_LTY/homework/Final Report/data/laptop.csv",
  na.strings = c("", "NA"))
```

Summary Statistics

```
latex(describe(laptop), file = "", caption.placement = "top")
```

laptop													
29 Variables							1020 Observations						
X													
n 1020	missing 0	distinct 1020	Info 1	Mean 509.5	Gmd 340.3	.05 50.95	.10 101.90	.25 254.75	.50 509.50	.75 764.25	.90 917.10	.95 968.05	
lowest : 0 1 2 3 4, highest: 1015 1016 1017 1018 1019													

Name

n	missing	distinct
1020	0	1020

lowest : Acer Aspire 3 14 A314-36M Laptop (Intel Core i3 N305/ 8GB/ 512GB SSD/ Win11 Home)

Acer Aspire 3 15 A31

highest: Zebronics Pro Series Z ZEB-NBC 3S 2023 Laptop (12th Gen Core i3 / 8GB/ 512GB SSD/ Win11 Home)

Zebronics Pro Series

Brand

n	missing	distinct
1020	0	31

lowest : Acer

Apple

Asus

ASUS

Avita

highest: Ultimius

Walker

Wings

Xiaomi

Zebronics

Price

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	561	1	82063	59619	25598	31990	43990	63690	94990	149990	199990

lowest : 8000 12990 13990 14490 14990, highest: 450990 453990 489883 519990 599990

Rating

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	17	0.996	4.374	0.2689	4.00	4.05	4.20	4.35	4.55	4.70	4.75

Value	3.95	4.00	4.05	4.10	4.15	4.20	4.25	4.30	4.35	4.40	4.45	4.50	4.55	4.60
Frequency	1	71	56	69	57	66	68	57	71	76	55	57	64	57
Proportion	0.001	0.070	0.055	0.068	0.056	0.065	0.067	0.056	0.070	0.075	0.054	0.056	0.063	0.056

Value	4.65	4.70	4.75
Frequency	53	64	78
Proportion	0.052	0.063	0.076

For the frequency table, variable is rounded to the nearest 0

Processor_brand

n	missing	distinct
1020	0	7

Value	AMD	Apple	HiSilicon	Intel	MediaTek	Microsoft	Qualcomm
Frequency	250	18	1	742	7	1	1
Proportion	0.245	0.018	0.001	0.727	0.007	0.001	0.001

Processor_name

n	missing	distinct
1020	0	36

lowest : AMD Athlon

AMD Athlon Pro

AMD Athlon Silver AMD Ryzen 3

AMD Ryzen 5

highest: Mediatek

MediaTek

MediaTek Kompanio Microsoft SQ1 SQ1 Qualcomm X Elite

Processor_variant

n	missing	distinct
996	24	125

lowest :	1005G1	100U	10210U	10210Y	1035G1
highest:	N5100	N6000	processor	SQ1	Z3735F

Processor_gen

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
891	129	13	0.95	10.45	3.086	5	5	7	12	13	13	13

Value		1	3	4	5	6	7	8	9	10	11	12	13	14
Frequency		1	11	4	92	17	120	7	1	13	113	246	255	11
Proportion		0.001	0.012	0.004	0.103	0.019	0.135	0.008	0.001	0.015	0.127	0.276	0.286	0.012

For the frequency table, variable is rounded to the nearest 0

Core_per_processor

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1008	12	12	0.977	8.572	4.694	2	4	6	8	10	14	16

Value		2	4	5	6	8	10	11	12	14	16	20	24
Frequency		81	118	8	182	198	189	1	93	69	42	3	24
Proportion		0.080	0.117	0.008	0.181	0.196	0.188	0.001	0.092	0.068	0.042	0.003	0.024

For the frequency table, variable is rounded to the nearest 0

Total_processor

n	missing	distinct	Info	Mean	Gmd
573	447	8	0.902	3.927	2.103

Value		1	2	4	5	6	8	10	12
Frequency		7	221	180	1	121	41	1	1
Proportion		0.012	0.386	0.314	0.002	0.211	0.072	0.002	0.002

For the frequency table, variable is rounded to the nearest 0

Execution_units

n	missing	distinct	Info	Mean	Gmd
573	447	5	0.734	6.998	2.478

Value		4	6	8	12	16
Frequency		193	3	350	3	24
Proportion		0.337	0.005	0.611	0.005	0.042

For the frequency table, variable is rounded to the nearest 0

Low_Power_Cores

n	missing	distinct	Info	Mean	Gmd
1020	0	2	0.124	0.08627	0.1653

Value		0	2
Frequency		976	44
Proportion		0.957	0.043

Energy_Efficient_Units

n	missing	distinct	Info	Sum	Mean	Gmd
1020	0	2	0.124	44	0.04314	0.08263

Threads

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
972	48	14	0.937	12.82	5.988	4	8	8	12	16	20	22

Value	2	4	5	6	8	12	14	16	18	20	22	24	28	32
Frequency	36	50	1	7	178	346	6	218	11	51	27	15	3	23
Proportion	0.037	0.051	0.001	0.007	0.183	0.356	0.006	0.224	0.011	0.052	0.028	0.015	0.003	0.024

For the frequency table, variable is rounded to the nearest 0

RAM_GB

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	10	0.798	13.99	6.415	8	8	8	16	16	16	32

Value	2	4	8	12	16	18	32	36	48	64
Frequency	1	25	377	2	543	3	62	1	1	5
Proportion	0.001	0.025	0.370	0.002	0.532	0.003	0.061	0.001	0.001	0.005

For the frequency table, variable is rounded to the nearest 0

RAM_type

n	missing	distinct
998	22	12

Value	DDR3	DDR4	DDR5	DDR6	LPDDR3	LPDDR4	LPDDR4X	LPDDR5	LPDDR5X	LPDDR4X
Frequency	2	522	188	1	1	16	53	173	38	2
Proportion	0.002	0.523	0.188	0.001	0.001	0.016	0.053	0.173	0.038	0.002

Value	PDDR5X	Unified
Frequency	1	1
Proportion	0.001	0.001

Storage_capacity_GB

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	10	0.664	627.7	265	256	512	512	512	512	1000	1000

Value	32	64	128	256	512	1000	1256	1512	2000	4000
Frequency	1	9	19	46	702	218	4	2	18	1
Proportion	0.001	0.009	0.019	0.045	0.688	0.214	0.004	0.002	0.018	0.001

For the frequency table, variable is rounded to the nearest 0

Storage_type

n	missing	distinct
1020	0	4

Value	Hard Disk	NVMe SSD	SSD	Hard Disk & SSD
Frequency	14	1	999	6
Proportion	0.014	0.001	0.979	0.006

Graphics_name

n	missing	distinct
1018	2	136

lowest :	10-Core GPU	10 Core GPU	14 Core GPU	18 Core GPU	30 Core GPU
highest:	Nvidia RTX 3050	NVIDIA RTX 4050	NVIDIA RTX A500	NVIDIA RTX NVIDIA T550	Radeon Pro 5500M

Graphics_brand

n	missing	distinct
1018	2	7

Value	Adreno	AMD	Apple	ARM	Intel	NVIDIA	Radeon
Frequency	1	151	18	7	486	354	1
Proportion	0.001	0.148	0.018	0.007	0.477	0.348	0.001

Graphics_GB

n	missing	distinct	Info	Mean	Gmd
368	652	6	0.871	5.94	2.556

Value	2	4	6	8	12	16
Frequency	4	174	87	80	12	11
Proportion	0.011	0.473	0.236	0.217	0.033	0.030

For the frequency table, variable is rounded to the nearest 0

Graphics_integreted

n	missing	distinct
1018	2	2

Value	False	True
Frequency	772	246
Proportion	0.758	0.242

Display_size_inches

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	22	0.869	15.16	1.008	13.5	14.0	14.0	15.6	15.6	16.0	16.1

lowest : 11.6 12 12.4 13 13.3, highest: 16.1 16.2 17 17.3 18

Horizontal_pixel

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	22	0.569	2036	340.2	1366	1920	1920	1920	1920	2560	2880

lowest : 1080 1200 1280 1366 1440, highest: 3024 3072 3200 3456 3840

Vertical_pixel

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	22	0.767	1214	272.7	768	1080	1080	1080	1200	1800	1864

lowest : 768 1024 1080 1200 1280, highest: 2000 2160 2234 2400 2560

ppi

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	62	0.902	157.2	30.93	127.3	141.2	141.2	141.2	161.7	212.3	242.6

lowest : 100.45 111.14 111.94 127.34 129.58, highest: 266.26 266.37 267.08 283.02 337.93

Touch_screen

n	missing	distinct
1020	0	2

Value	False	True
Frequency	904	116
Proportion	0.886	0.114

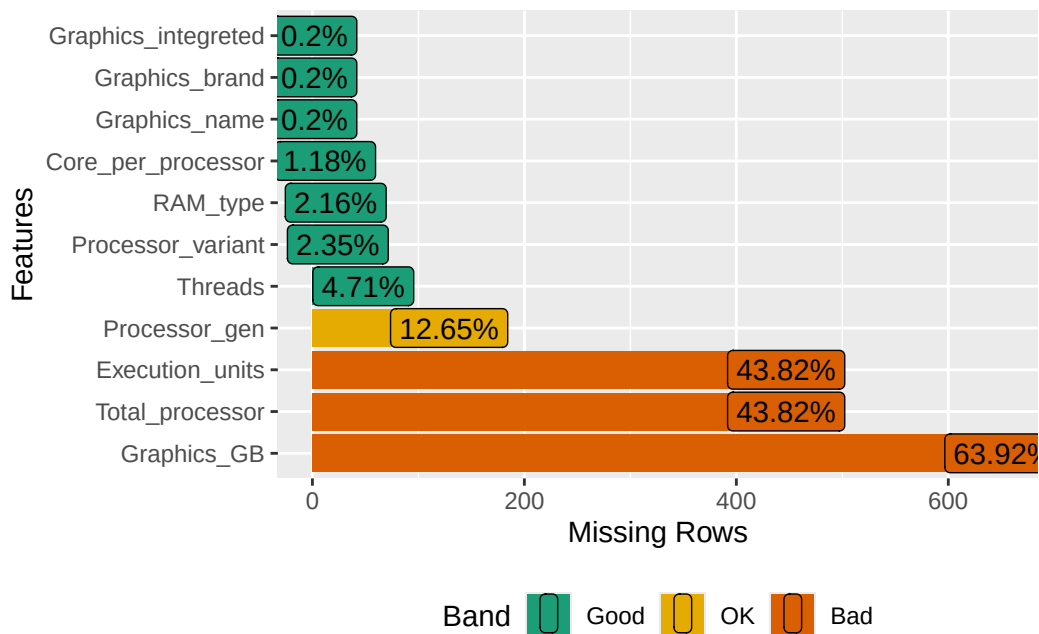
Operating_system

n	missing	distinct
1020	0	13

lowest :	Android 11 OS	Chrome OS	DOS 3.0 OS	DOS OS	jio
highest:	Mac OS	Prime OS	Ubuntu OS	Windows 10 OS	Windows 11 OS

Missing Values

```
plot_missing(laptop, missing_only = TRUE)
```



1. **Good:**

- “Graphics_integrated”, “Graphics_brand”, “Graphics_name” have **2(0.2%)** missing values.
- “Core_per_processor” has **12(1.18%)** missing values.
- “RAM_type” has **22(2.16%)** missing values.
- “Processor_variant” has **24(2.35%)** missing values.
- “Threads” has **48(4.71%)** missing values.

OK:

- “Processor_gen” has **129(12.65%)** missing values.
- “Execution_units” and “Total_processor” have **447(43.82%)** missing values.

Bad:

- “Graphics_GB” has **652(63.92%)** missing values

Data Visualization


```
laptop[laptop$Brand == "Asus", "Brand"] = "ASUS"
sort(table(laptop$Brand))
```

Colorful	iBall	Jio	Ninkear	Razer	Tecno	Walker	AXL
1	1	1	1	1	1	1	2
Huawei	Fujitsu	Primebook	Wings	Ultimus	Avita	Gigabyte	Honor
2	3	3	3	5	6	6	6
Xiaomi	LG	Microsoft	Zebronics	Chuwi	Apple	Infinix	Samsung
6	7	7	7	8	20	20	32
Acer	MSI	Dell	ASUS	HP	Lenovo		
69	97	116	158	213	217		

Data Analysis

Feature Engineering

```
data <- laptop
```

Split training and testing dataset

```
set.seed(2024)
# smp.size = floor(0.8*nrow(data))
# train.index = sample(seq_len(nrow(data)), smp.size)
# train = data[train.index, ] # 80%
# test = data[-train.index, ] # 20%
```

Model Fitting

Linear Regression

SVM Regressor

```
library(e1071, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
# model <- svm(y ~ x , data)
# svr.pred = predict(model, data)
# mean(abs(data$y - svr.pred))
```

Random Forest Regressor

```
library(randomForest, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
```

XGBoost

The parameters of XGBoost are listed below: max.depth, eta, nthread, nrounds, objective: “binary:logistic”, verbose: Valid values are 0 (silent), 1 (warning), 2 (info), 3 (debug).

```
library(xgboost, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
# xgb <- xgboost(data = train, max.depth = 2, eta = 1, nthread = 2, nrounds = 2, objective =
```

LightGBM

```
library(lightgbm, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
# lgbm <- lightgbm()
```

Catboost

```
library(catboost, warn.conflicts = FALSE, verbose = FALSE, quietly = TRUE)
```

Final Results